MSG No. 125.364 $P4/nbm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4221'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 2: Wyckoff site: 2b, site symmetry: 4221'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 3: Wyckoff site: 2c, site symmetry: -42m1'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[1,4,5,6,10,11,15,16,17,20,21,22,26,27,31,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[2,3,7,8,9,12,13,14,18,19,23,24,25,28,29,30]

Table 4: Wyckoff site: 2d, site symmetry: -42m1'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,4,5,6,10,11,15,16,17,20,21,22,26,27,31,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,3,7,8,9,12,13,14,18,19,23,24,25,28,29,30]

Table 5: Wyckoff site: 4e, site symmetry: ..2/m1'

No.	position	mapping
1	$[0, 0, 0]$	[1,7,9,15,17,23,25,31]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,5,10,13,18,21,26,29]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,4,11,12,19,20,27,28]
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,8,14,16,22,24,30,32]

Table 6: Wyckoff site: 4f, site symmetry: ..2/m1'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,7,9,15,17,23,25,31]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,5,10,13,18,21,26,29]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,11,12,19,20,27,28]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[6,8,14,16,22,24,30,32]

Table 7: Wyckoff site: `4g`, site symmetry: `4..1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,3,6,17,18,19,22]`
2	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[4,5,7,8,20,21,23,24]`
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9,10,11,14,25,26,27,30]`
4	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[12,13,15,16,28,29,31,32]`

Table 8: Wyckoff site: `4h`, site symmetry: `2.mm1'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,6,15,16,17,22,31,32]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[2,3,12,13,18,19,28,29]`
3	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[4,5,10,11,20,21,26,27]`
4	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7,8,9,14,23,24,25,30]`

Table 9: Wyckoff site: `8i`, site symmetry: `..21'`

No.	position	mapping
1	$[x, x, 0]$	`[1,7,17,23]`
2	$\left[\frac{1}{2} - x, x, 0\right]$	`[2,5,18,21]`
3	$\left[x, \frac{1}{2} - x, 0\right]$	`[3,4,19,20]`
4	$\left[\frac{1}{2} - x, \frac{1}{2} - x, 0\right]$	`[6,8,22,24]`
5	$[-x, -x, 0]$	`[9,15,25,31]`
6	$\left[x + \frac{1}{2}, -x, 0\right]$	`[10,13,26,29]`
7	$\left[-x, x + \frac{1}{2}, 0\right]$	`[11,12,27,28]`
8	$\left[x + \frac{1}{2}, x + \frac{1}{2}, 0\right]$	`[14,16,30,32]`

Table 10: Wyckoff site: `8j`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x, x, \frac{1}{2}\right]$	`[1,7,17,23]`
2	$\left[\frac{1}{2} - x, x, \frac{1}{2}\right]$	`[2,5,18,21]`
3	$\left[x, \frac{1}{2} - x, \frac{1}{2}\right]$	`[3,4,19,20]`
4	$\left[\frac{1}{2} - x, \frac{1}{2} - x, \frac{1}{2}\right]$	`[6,8,22,24]`
5	$\left[-x, -x, \frac{1}{2}\right]$	`[9,15,25,31]`
6	$\left[x + \frac{1}{2}, -x, \frac{1}{2}\right]$	`[10,13,26,29]`
7	$\left[-x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,27,28]`
8	$\left[x + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	`[14,16,30,32]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x, \frac{1}{4}, 0\right]$	`[1,4,17,20]`
2	$\left[\frac{1}{4}, x, 0\right]$	`[2,7,18,23]`

continued ...

Table 11

No.	position	mapping
3	$\left[\frac{1}{4}, \frac{1}{2}-x, 0\right]$	[3,8,19,24]
4	$\left[\frac{1}{2}-x, \frac{1}{4}, 0\right]$	[5,6,21,22]
5	$\left[-x, \frac{3}{4}, 0\right]$	[9,12,25,28]
6	$\left[\frac{3}{4}, -x, 0\right]$	[10,15,26,31]
7	$\left[\frac{3}{4}, x+\frac{1}{2}, 0\right]$	[11,16,27,32]
8	$\left[x+\frac{1}{2}, \frac{3}{4}, 0\right]$	[13,14,29,30]

Table 12: Wyckoff site: 8l, site symmetry: .2.1'

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{2}\right]$	[1,4,17,20]
2	$\left[\frac{1}{4}, x, \frac{1}{2}\right]$	[2,7,18,23]
3	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{2}\right]$	[3,8,19,24]
4	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{2}\right]$	[5,6,21,22]
5	$\left[-x, \frac{3}{4}, \frac{1}{2}\right]$	[9,12,25,28]
6	$\left[\frac{3}{4}, -x, \frac{1}{2}\right]$	[10,15,26,31]
7	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{2}\right]$	[11,16,27,32]
8	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	[13,14,29,30]

Table 13: Wyckoff site: 8m, site symmetry: ..m1'

No.	position	mapping
1	$[x, -x, z]$	[1,15,17,31]
2	$\left[x+\frac{1}{2}, x, z\right]$	[2,13,18,29]
3	$\left[-x, \frac{1}{2}-x, z\right]$	[3,12,19,28]
4	$\left[x, x+\frac{1}{2}, -z\right]$	[4,11,20,27]
5	$\left[\frac{1}{2}-x, -x, -z\right]$	[5,10,21,26]
6	$\left[\frac{1}{2}-x, x+\frac{1}{2}, z\right]$	[6,16,22,32]
7	$[-x, x, -z]$	[7,9,23,25]
8	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -z\right]$	[8,14,24,30]

Table 14: Wyckoff site: 16n, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,17]
2	$\left[\frac{1}{2}-y, x, z\right]$	[2,18]
3	$\left[y, \frac{1}{2}-x, z\right]$	[3,19]
4	$\left[x, \frac{1}{2}-y, -z\right]$	[4,20]
5	$\left[\frac{1}{2}-x, y, -z\right]$	[5,21]
6	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	[6,22]
7	$[y, x, -z]$	[7,23]
8	$\left[\frac{1}{2}-y, \frac{1}{2}-x, -z\right]$	[8,24]
9	$[-x, -y, -z]$	[9,25]

continued ...

Table 14

No.	position	mapping
10	$\left[y+\frac{1}{2},\,-x,\,-z\right]$	[10,26]
11	$\left[-y,\,x+\frac{1}{2},\,-z\right]$	[11,27]
12	$\left[-x,\,y+\frac{1}{2},\,z\right]$	[12,28]
13	$\left[x+\frac{1}{2},\,-y,\,z\right]$	[13,29]
14	$\left[x+\frac{1}{2},\,y+\frac{1}{2},\,-z\right]$	[14,30]
15	$[-y,\,-x,\,z]$	[15,31]
16	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,z\right]$	[16,32]